Gaohan (Elijah) Ye

Latest Update: 10/23/2026

CONTACT

3614 Flora Vista Ave., APT 273
Santa Clara, CA 95051

Email: elijah.gaohan.ye@gmail.com
Website: elijah-ye.github.io/
Phone: (510) 520-7207
LinkedIn: linkedin.com/in/elijahye/

TECHNICAL SKILLS

Programming: SystemVerilog (Advanced), C/C++ (Advanced), UVM/OVM, Python, Tcl, Bash, CUDA, Assembly (x86, RISC-V)
Hardware Design: RTL Design, CPU Architecture, RISC-V, x86, CXL/PCIe, Cache Design, Pipeline Optimization
Physical Design: Custom CMOS Layout, Standard Cell Design, Transistor Sizing, Floorplanning, DRC/LVS Verification
Hardware Verification: Design Verification, UVM/OVM, Functional & Toggle Coverage, SystemVerilog Assertions, Test Plans, System-Level Testing, FuSa/Error Injection
DFT: ATPG Pattern Validation, Boundary Scan (JTAG/BSDL), MBIST, STIL Patterns, Gate-Level Simulation, Die-to-Package Test Collateral
Technologies/Tools: Cadence Virtuoso, Innovus, Calibre (DRC/LVS), Synopsys TestMAX ATPG, Verdi, VCS, Design Compiler, Git, Linux, Slurm, Intel Extension for PyTorch, Intel MLC

WORK EXPERIENCE

Astera Labs Jun 2026 – Oct 2026
DFT Engineer Intern, Leo CXL Memory Controller San Jose, CA

- Debugged transition-delay (TDF) ATPG silicon failures on I3C/I2C logic: found the patterns had been generated with no timing exceptions loaded (an SDC read error), then wrote false-path constraints taking uncovered I3C paths from 103 to 0
- Inserted Synopsys STAR Memory System (SMS) BIST/repair into 3 blocks covering 186 SRAM macros in 9 vendor configurations
- Root-caused two MBIST clock-failure classes: a BIST clock held off by reset (5/5 with a testbench force) and a two-clock-domain topology (9/9 BIST processors by configuration alone)
- Ran a full DFT gate-level release regression for another chip's B0 stepping (25/26 MBIST, 9/9 boundary-scan at 0 mismatches), tracing the one failure to netlists that carried the previous stepping's memory BIST
- Found why a serial-TAP MBIST test had never passed on the new stepping: power-up BIST held the memory BIST clock on a 10 MHz reference instead of TCK
- Traced a 2.4M-mismatch all-X boundary-scan simulation to 10 TAP/control pins typed as inputs, causing TDI pad contention; fixed with BSDL retyping, a reset hold, and a flush preamble
- Root-caused a 64-mismatch AC-input boundary-scan failure to the pattern tool over-predicting expects on 16 observe-only cells, reaching 0 mismatches on all 9 pattern types for DDR4 and DDR5
- Built two Python toolchains (∼4,000 lines) with byte-exact regression checks, automating die-to-package pin remapping of boundary-scan BSDL and ATE STIL patterns and replacing manual bash/dc_shell workflows
- Verified converted patterns end-to-end with Slurm-dispatched VCS gate-level simulation against the synthesized netlist, debugging failures in Verdi
- Generated block-level stuck-at ATPG patterns for 2 blocks and ported them to a top-level testbench: 205 patterns, 0 mismatches

Rivian Automotive May 2025 – Aug 2025
Design Verification Intern, RAP1 Silicon Team Champaign, IL

- Developed system-level testcases for RAP1 (Rivian Autonomy Processor), a custom 5nm AI inference chip delivering 1600 sparse TOPS for autonomous driving compute
- Increased toggle coverage to over 90% via a new, more efficient coverage generation flow
- Enhanced an existing integration test suite, reducing the failure rate from over 40% to less than 5%
- Developed a new system-level test to validate memory access, creating a baseline for future verification tests
- Authored new FuSa tests to validate error injection and interrupt mechanisms for critical components

University of Illinois at Urbana-Champaign Jan 2025 – May 2026
Teaching Assistant for Digital Systems Laboratory, ECE 385 Champaign, IL

- Evaluated SystemVerilog/FPGA projects and microprocessor system implementations
- Conducted technical demonstrations and assessments of digital circuits, state machines, and SoC designs
- Graded lab reports covering combinational/sequential logic, timing analysis, and hardware-software co-design

Rivian Automotive Apr 2024 – Aug 2024
Design Verification Intern, RAP1 Silicon Team Champaign, IL

- Created performance monitor to track AXI transaction signals to test read and write operation performance
- Increased toggle coverage of chip blocks by creating and implementing new targeted tests
- Conducted system-level verification for the DUT using a custom UVM testbench
- Developed and executed detailed test plans, identifying bugs and successfully debugging and resolving issues

University of Illinois at Urbana-Champaign Jan 2023 – Dec 2024
Teaching Assistant for Computer System Engineering, ECE 391 Champaign, IL

- Led weekly lab and discussion sessions, teaching students about operating system concepts including process scheduling, virtual memory, file systems, and system call implementation
- Offered comprehensive support to students, addressing inquiries regarding x86, virtual memory, scheduling, file systems, and course materials
- Graded over 300 student assignments and conducted demos of their custom operating systems
- Collaborated with professors in the grading process for midterms and final exams

Headline (a Venture Capital Company based in San Francisco) May 2022 – Aug 2022
Frontend Developer Intern Remote

- Added display features and security auditing in a custom integration with Gmail that exposes several million highly sensitive emails across the organization by replying on Rails backend and React front end
- Implemented internal management tool with React.js; restructured API calls to improve page latency by 30%
- Improved performance of Streak CRM and Gmail integrations by fixing timeout bugs

Publications & Research

"Transparent Memory Management for Large-Scale LLM Training and Tuning" **2026**
(Under Review)
Advisor: *Nam Sung Kim* Dec 2024 – Present

- Characterized LLM training memory-access patterns, revealing strong spatial and cross-iteration temporal locality in large tensor accesses
- Findings motivated transparent UVM-based CPU-memory offloading with a coalescing prefetcher to relieve network bottlenecks in distributed training

"ReScue: Reliable and secure CXL memory" **2026**
Chihun Song, Austin Antony Cruz, Michael Jaemin Kim, Minbok Wi, **Gaohan Ye**, Kyungsan Kim, Sangyeol Lee, Jung Ho Ahn, and Nam Sung Kim
IEEE International Symposium on High-Performance Computer Architecture (HPCA), Jan 2026

"Exploiting Intel Advanced Matrix Extensions (AMX) for LLM Inference" **2024**
IEEE Computer Architecture Letters (CAL), 2024, IEEE Best Paper Award
Advisor: *Nam Sung Kim* Dec 2023 – Apr 2024

- Characterized Intel AMX GEMM/GEMV throughput on a dual-socket Sapphire Rapids server across OPT-125M to OPT-175B matrix shapes (32.7 TFLOPS peak GEMM; GEMV memory-bandwidth-bound)
- Established the CPU performance profile behind an arithmetic-intensity-based CPU-GPU partitioning policy; the team's system achieved 12.1× lower latency and 2.2× higher throughput than GPU-only OPT-30B inference

Relevant Projects

VLSI Physical Design | *Virtuoso, Innovus, Design Compiler* Jan 2026 – May 2026
Course Participant, FreePDK45 Champaign, IL

- Built an area-optimized custom CMOS standard cell library, verified via DRC, LVS, and digital simulation
- Designed schematic and physical layout of a bit-sliced datapath with a custom ALU and register file
- Optimized CMOS transistor sizing to balance strict area constraints with worst-case pull-up/pull-down drive strength equivalent to a minimum-size NMOS
- Ran Design Compiler synthesis and Innovus place-and-route, taking RTL to routed layout

NPU-CPU LLM Inference | *Furiosa NPU, Intel AMX* Aug 2025 – Dec 2025
Project Lead, ECE 527 Champaign, IL

- Designed an analytical performance model partitioning Llama 3.1 8B inference across a Furiosa RNGD NPU and an Intel AMX CPU by arithmetic intensity and batch size, predicting end-to-end latency from measured component latencies
- Derived three batch-size-adaptive policies; hybrid NPU prefill + CPU decode showed 12% average latency overhead vs. H100 at 2.6× lower system power
- Projected 23–40% lower three-year total cost of ownership than H100-based systems

RISC-V Processor | *SystemVerilog, RISC-V* Sept 2023 – Dec 2023
Course Participant Champaign, IL

- Designed and implemented RV32I processor in SystemVerilog with data & branch hazard detection as team
- Achieved 28.6% frequency increase and 47% cache stall reduction, securing 3rd place among 30 groups
- Applied timing analysis and logic optimization principles for hardware-software co-design

Linux Kernel | *C, x86* Oct 2022 – Dec 2022
Course Participant Champaign, IL

- Created a Linux Kernel featuring 3 terminals and 6 processes
- Implemented read-only File System, Round-Robin Scheduling, 4kB and 4mB Paging
- Successfully handled interrupts, exceptions, and system calls from user programs

FPGA Flappy Bird Game | *SystemVerilog, Python, C* Jan 2022 – May 2022
Course Participant Champaign, IL

- Programmed FPGA to output a VGA signal to draw game visuals to a monitor
- Enabled keyboard-FPGA communication via SPI to run C-based keyboard drivers on SoC
- Wrote Python scripts to compress PNG format into MIF format to instantiate onto FPGA on-chip memory
- Implemented state machine in SystemVerilog to animate walking character depending on keyboard inputs
- Detected collisions between map and player with state machine-controlled memory address into ROMs

Education

University of Illinois Urbana-Champaign Aug 2024 – May 2026
Master of Science, Electrical and Computer Engineering GPA: 3.86/4.00

University of Illinois Urbana-Champaign Aug 2020 – Dec 2023
Bachelor of Science, Computer Engineering GPA: 3.96/4.00

Relevant Coursework
Computer Architecture, SoC/Hardware Design, VLSI System Design, Network Protocol, Operating System, Data Structure, Algorithm

Awards and Distinctions

IEEE Best Paper Award **2024**

A.R. "Buck" Knight Scholarship, ECE Department **Aug 2023**

Oakley Award in Electrical and Computer Engineering **Mar 2023**

Bradley A. Simmons Memorial Scholarship, ECE Department **Sep 2022**